MIL-I-12058C
27 October 1982
SUPERSEDING
MIL-I-12058B
12 November 1969

MILITARY SPECIFICATION

IRON, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of iron powder.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test
PPP-C-96 - Cans, Metal, 28 Gage and Lighter

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193 - Reagent Water
E50 - Apparatus, Reagents, and Safety Precautions for Chemical Analysis of Metals

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical characteristics. Iron shall conform to the chemical characteristics of table I when tested as specified therein.

3.2 Particle size characteristics. Iron shall conform to the particle size characteristics of table II when tested as specified in 4.2.4.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified

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TABLE I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Total iron	97.0	--	4.2.4.1
Manganese	--	0.10	4.2.4.2
Metallic iron	89.0	--	4.2.4.3
Grease and oil	--	0.05	4.2.4.4
Carbon	--	0.10	4.2.4.5
Silicon	--	0.10	4.2.4.6

TABLE II. Particle size characteristics

Sieve size	Percent by weight passing	
	Minimum	Maximum
No. 100	99.9	--
No. 200	75.0	--
No. 325	25.0	50.0

in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the iron produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For iron test. Sampling shall be conducted in accordance with table III. A representative specimen of approximately 500 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

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100 ml with water. Pipet 5, 10, and 20 ml of this solution into separate 100-ml volumetric flasks. Dilute each flask to 100 ml with water. (The three standard solutions contain 5, 10, and 20 micrograms of iron per milliliter, respectively.) Commercially prepared standard iron solutions having identical concentrations may also be used.

(b) Specimen test solution. Weigh to the nearest 0.1 milligram (mg) approximately 0.5 g of the specimen into a 250-ml Erlenmeyer flask. Add 1 to 1 hydrochloric acid until the specimen is wet. Gently heat the wetted specimen to start reaction and continue heating with care while adding the minimum amount of 1 to 1 hydrochloric acid necessary to dissolve the specimen. Quantitatively transfer the solution to a 500-ml volumetric flask and dilute to the mark with water. This is solution A. (Retain this solution for use in 4.2.4.2.) Pipet 10 ml of solution A into a 100-ml volumetric flask and dilute to the mark with water. Pipet 10 ml of the diluted solution into a 100-ml volumetric flask and dilute to the mark with water. This is the specimen test solution.

(c) Atomic absorption spectrophotometer conditions. Use the following atomic absorption spectrophotometer conditions:

(1) Burner - triflame, 10-centimeter groove
(2) Fuel - acetylene (oxidant - air)
(3) Wave length - 372.0 nanometers

(d) Procedure. Operate the atomic absorption spectrophotometer according to the manufacturer's instructions. Aspirate water into the flame for the zero setting (absorption or concentration mode setting depending on the instrument used). Aspirate the three standard solutions prepared as specified in (a) and record the absorbance or concentration reading of each standard solution, depending on the mode selected. Plot a calibration curve from the readings. Aspirate the specimen test solution prepared as specified in (b) and record the absorbance or concentration reading. Read the iron concentration of the specimen test solution in micrograms per milliliter from the calibration curve. Calculate the percent by weight total iron as follows:

$$\text{Percent by weight total iron} = \frac{5000\ A}{B}$$

where: A = Iron concentration in micrograms per milliliter of specimen test solution, and
B = Weight in milligrams of original specimen.

4.2.4.2 Manganese.

(a) Standard solutions. Prepare standard solutions having manganese contents of 10, 5, and 2.5 micrograms per milliliter, respectively, in water.

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(b) Atomic absorption spectrophotometer conditions. Use the following atomic absorption spectrophotometer conditions:

(1) Burner - triflame, 10-centimeter groove
(2) Fuel - acetylene (oxidant - air)
(3) Wave length - 280.1 nanometers
(4) Hollow cathode tube - manganese

(c) Procedure. Operate the atomic absorption spectrophotometer according to the manufacturer's instructions. Aspirate water into the flame for the zero setting (absorption or concentration mode setting depending on the instrument used). Aspirate the three standard solutions prepared as specified in (a) and record the absorbance or concentration reading of each standard solution, depending on the mode selected. Plot a calibration curve from the readings. Aspirate solution A from 4.2.4.1(b) and record the absorbance or concentration reading. Read the manganese concentration of solution A in micrograms per milliliter from the calibration curve. Calculate the percent by weight manganese as follows:

$$\text{Percent by weight manganese} = \frac{50\ A}{B}$$

where: A = Manganese concentration in micrograms per milliliter of solution A, and
B = Weight in milligrams of original specimen in 4.2.4.1.

4.2.4.3 Metallic iron.

(a) Manganous sulfate solution. Dissolve 67 g of manganous sulfate, tetrahydrate ($MnSO_4 \cdot 4H_2O$) in 500 to 600 ml of water. Add 138 ml of phosphoric acid having a specific gravity of 1.7, and 130 ml of concentrated sulfuric acid. Dilute to 1 liter with water.

(b) Procedure. Weigh to the nearest 0.1 mg approximately 0.5 g of the specimen into a 100-ml volumetric flask that has been previously filled with carbon dioxide. Dissolve 2.5 g of anhydrous ferric chloride ($FeCl_3$) in 50 ml of cold water and add this solution to the flask. (The ferric chloride should give a clear solution in cold water and be free from ferrous salt.) Displace any air in the flask with carbon dioxide, stopper immediately, and shake for 15 minutes. Dilute to the mark with water and mix. Pour the contents of the flask through filter paper, discard the first 15 ml, and catch the rest of the filtrate in a flask filled with carbon dioxide. Remove a 20-ml aliquot of the filtrate with a pipet previously filled with carbon dioxide and place the aliquot in a 300-ml Erlenmeyer flask previously filled with carbon dioxide. Add 20 ml of 7N sulfuric acid and 10 ml of manganese sulfate solution prepared as specified in (a). Dilute to 200 ml with water and titrate with 0.1N potassium permanganate solution to a faintly pink end point. Calculate the percent by weight metallic iron as follows:

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$$\text{Percent by weight metallic iron} = \frac{9.31\ AB}{W}$$

where: A = Milliliters of potassium permanganate solution used,
B = Normality of potassium permanganate solution, and
W = Weight in grams of specimen.

4.2.4.4 Grease and oil. Dry a 150-ml beaker at 87° to 93°C for 1 hour. Cool the beaker to room temperature in a desiccator and weigh to the nearest milligram. Weigh to the nearest 0.01 g approximately 30 g of the specimen and transfer to the extraction thimble of a large Soxhlet extraction apparatus. Place the thimble in the apparatus. Using 250 ml of anhydrous diethyl ether as the extraction liquid, extract the specimen for 1 hour. Evaporate the ether extract in the extraction flask to approximately 50 ml on a steam bath and transfer to the weighed beaker. Rinse the extraction flask with three 10-ml portions of ether and add to the extract in the beaker. Cover the beaker with a ribbed watchglass. After the solvent is removed, heat the beaker and residue at 87° to 93°C for 1 hour. Cool the beaker and residue to room temperature in a desiccator and weigh to the nearest milligram. Retain the extracted specimen for use in 4.2.4.5. Calculate the percent by weight grease and oil as follows:

$$\text{Percent by weight grease and oil} = \frac{100\ (A - B)}{W}$$

where: A = Weight in grams of beaker and residue,
B = Weight in grams of beaker, and
W = Weight in grams of specimen.

4.2.4.5 Carbon. Assemble a combustion train apparatus for determination of carbon by the direct-combustion method as shown in ASTM E50. Fill the absorbing bottles and towers with the proper reagents. Test the apparatus by heating the combustion tube and passing oxygen through the train. Heat the combustion furnace to 1100°C or above while passing oxygen through the train at about 150 ml per minute. After about 1/2 hour, remove the absorbing tube containing "Ascarite" and weigh to the nearest 0.1 mg against an extra absorbing tube which is similarly filled with "Ascarite" and oxygen, and is used as a couterpoise. Make a small groove in the alundum lining of the combustion boat and transfer approximately 1.75 g of the ether extracted specimen from 4.2.4.4, weighed to the nearest milligram, to the groove, distributing it evenly. Add 1 pellet of tin shot, about 3 millimeters in diameter, to serve as a catalyst. Cover the specimen with the boat cover and insert the covered boat into the hot combustion tube. Close the tube and allow the specimen to heat for 1 to 2 minutes, depending on the size of the particles. Admit oxygen gas at a rate of 800 to 1000 ml per minute while the combustion is going on. When combustion is complete (after about 1.5 to 2 minutes), reduce the rate of flow of oxygen to about 150 ml per minute and continue for 6 to 8 minutes to sweep out the carbon dioxide. Remove the carbon dioxide absorption tube filled with oxygen, place it in the balance case for 10 minutes, open momentarily, and weigh against a similar tube used as a counterpoise. The increase in weight represents carbon

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dioxide. Remove the boat from the tube and examine the fusion for evidence of incomplete combustion. If the powder is not thoroughly fused in a solid pig, reject the determination. The carbon dioxide tube will warm up when carbon dioxide is absorbed. It is not necessary to wait until it reaches room temperature before weighing it if it is in continuous use, provided the same time interval is maintained and approximately the same amount of carbon dioxide is absorbed. Calculate the percent by weight carbon as follows:

$$\text{Percent by weight carbon} = \frac{27.29\ A}{W}$$

where: A = Weight in grams of carbon dioxide obtained, and
W = Weight in grams of specimen.

4.2.4.6 Silicon. Weigh to the nearest 0.1 mg approximately 5 g of the specimen and transfer to a 400-ml beaker. Add 30 ml of concentrated hydrochloric acid and heat to dissolve. Cool the solution, then cautiously, under a hood, add 30 ml of 1 to 1 nitric acid and 25 ml of concentrated perchloric acid. Care should be observed in the handling and heating of perchloric acid. Evaporate until copious white fumes are observed and fume for 15 minutes. Cool, cautiously add 150 ml of hot water, stir well, and heat until salts are dissolved. Add shredded Whatman No. 40 filter paper and filter through another piece of the same filter paper. Wash the paper and residue thoroughly with hot 1 to 99 sulfuric acid. Place the paper and washed residue in a platinum crucible. Dry and heat carefully until free of carbon, and then ignite over a Meker burner for 25 minutes. Cool in a desiccator, weigh to the nearest 0.1 mg, and repeat the ignition until constant weight is reached. Moisten the residue of impure dioxide with a few drops of water. Add 1 to 2 drops of concentrated sulfuric acid. Under a hood, cautiously add about 5 ml of concentrated hydrofluoric acid and evaporate to dryness. Ignite over a Meker burner, cool in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent by weight silicon as follows:

$$\text{Percent by weight silicon} = \frac{46.72\ A}{W}$$

where: A = Weight in grams of silicon dioxide in specimen (loss in weight after hydrofluoric acid treatment) and
W = Weight in grams of specimen.

4.2.4.7 Particle size. Use sieves conforming to RR-S-366 of the sizes specified in 3.2. Weigh the dry sieves to the nearest 0.1 g and nest them in order of increasing fineness with the coarest sieve on top, so that the material passing through a sieve is transferred directly to the net sieve in the series. Place the sieves on a weighed bottom pan. Weigh to the nearest 0.1 g approximately 100 g of the specimen and transfer to the first (coarsest) sieve, and then cover the assembly. Shake for 5 minutes by means of a mechanical shaker geared to produce 300 $\pm$ 15 gyrations and 150 $\pm$ 10 taps of the striker per minute. Record the final weight of the residue retained on each sieve and calculate the percent by weight specimen passing through each sieve.

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4.2.5 Container leakage test. Place the container in each of the following positions, and leave it in each for a period of 15 minutes:

(a) Upright
(b) Upside down
(c) On one side (or one quadrant)
(d) On one end (or second quadrant)
(e) On other side (or fourth quadrant)

Examine the container after each period for any evidence of leakag.

5. PACKAGING

5.1 Unit packing, level C. Iron powder shall be unit packed level C in a 25- or 500-pound (lb) quantity as specified (see 6.2).

5.1.1 Twenty-five-pound quantity. A quantity of 25 (+0.25 or -0) lb of iron powder shall be unit packed in a clean, dry nominal 1-gallon (gal) capacity metal can. The can shall conform to type V, class 2 of PPP-C-96, and shall be closed as specified in PPP-C-96.

5.1.2 Five-hundred-pound quantity. A quantity of 500 (+0.5 or -0) lb of iron powder shall be unit packed in a clean, dry nominal 20-gal steel drum. The drum shall conform to Rule 40 of the Uniform Freight Classification Rules. The drum shall be closed tightly and shall not show any evidence of leakage of contents when tested as specified in 4.2.5.

5.2 Packing.

5.2.1 Twenty-five pound quantity. Four cans of iron powder, unit packed as specified in 5.1.1, shall be packed in a close-fitting fiberboard or nailed wood box, or wirebound wood box, in compliance with Uniform Freight Classification Rules, National Motor Freight Classification Rules or other carrier rules as applicable to the mode of transportation. Containers shall be capable of being stacked and supporting superimposed loads during shipment, and shall assure safe delivery to destination without damage to contents.

5.2.2 Five-hundred pound quantity. The 500-lb quantity, unit packed as specified in 5.1.2, shall require no further protection for shipment.

5.2.3 Unitiation. Uniform quantities of iron powder packs shall be palletized in accordance with the applicable requirements of MIL-STD-147, and in a manner to assure acceptance by common carrier, safe delivery to destination and stability in stacking and storage.

5.4 Marking. Containers and pallet loads shall be marked in accordance with MIL-STD-129 and any applicable regulatory requirements. Marking shall include the lot or batch number and date of manufacture of the iron powder.

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6. NOTES

6.1 Intended use. Iron is intended for use in pyrotechnic mixtures.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification and the unit quantity of iron required (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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